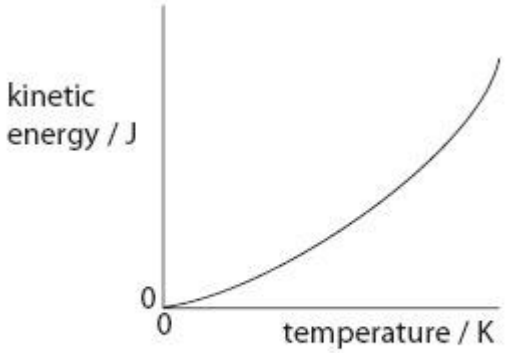
Questions

Q1.

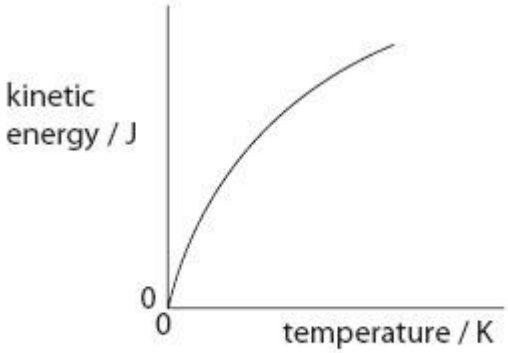
(a) Which graph shows the way in which the average kinetic energy of the molecules of a gas changes with temperature?

Put a cross (☒) in the box next to your answer.

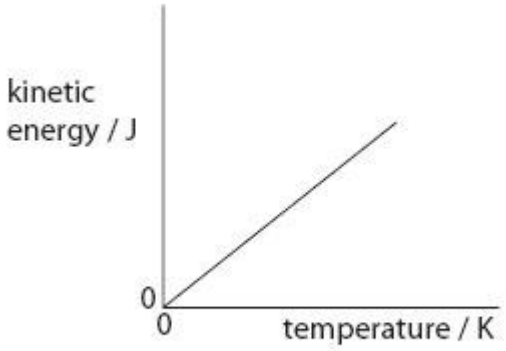
(1)



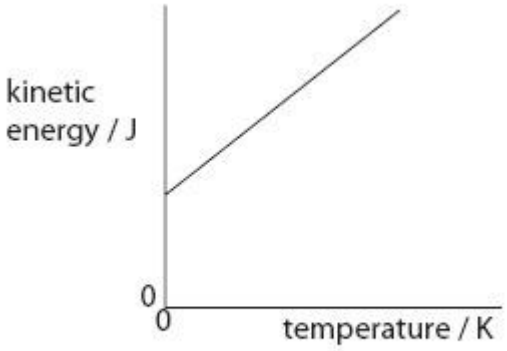
☐ **A**



☐ **B**



☐ **C**



☐ **D**

(b) The photograph shows a scuba diver.
She can breathe under water because she carries a cylinder of air on her back.

(i) The air molecules in the cylinder move randomly.
Describe how these air molecules exert a pressure on the cylinder.

(2)

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(ii) The cylinder contains air at a pressure of 21 000 kPa.

The volume of air in the cylinder is 15.0 litres.
When the valve on the cylinder is opened, the air expands until its pressure is 100 kPa.
The temperature of the air does not change.
Show that the new volume of air is about 3 200 litres.

(2)

(iii) The cylinder is filled with air in a hot country and then taken to a cold country.

The temperature in the hot country is 305 K.
The temperature in the cold country is 278 K.
The pressure in the cylinder in the hot country is 21 000 kPa.
Calculate the pressure in the cylinder in the cold country.

(3)

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(Total for Question = 8 marks)

Q2.

*A container is sealed so that the mass of the gas inside cannot change.

The volume of the gas is changed and the pressure is measured at different volumes.

The temperature of the gas does not change.

Figure 18 is a graph of the results.

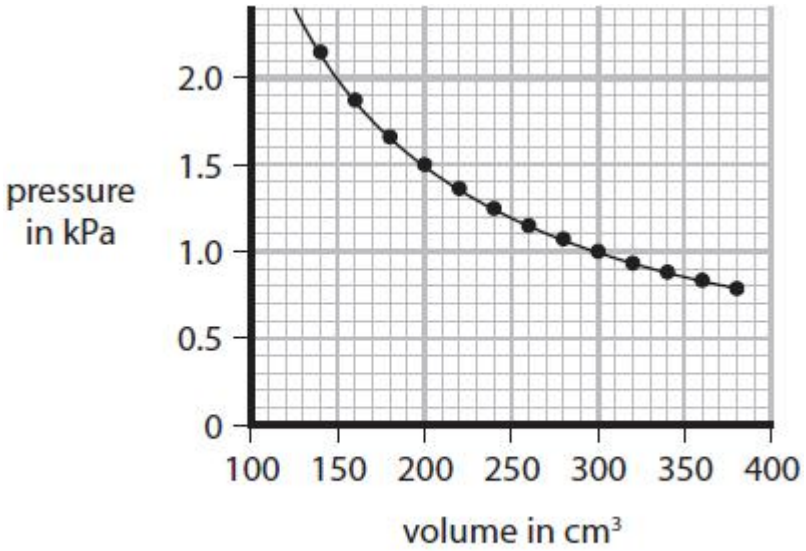

Figure 18

Explain, in terms of the movement of particles, why there is a pressure on the container and why the pressure changes as shown in Figure 18.

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(Total for question = 6 marks)

Q3.

Some gas is trapped in a container similar to the container in Figure 19a.

The gas is compressed at a constant temperature.

Explain, **in terms of gas particles**, why the pressure of the gas increases when the volume decreases.

Your answer should refer to

- how the gas particles exert a pressure
- why the pressure increases when the volume decreases.

(6)

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(Total for question = 6 marks)

Q4.

Figure 16 shows a cylinder containing some gas.

A piston prevents the gas escaping.

In Figure 16x a small weight pushes down and the volume and pressure of the gas are measured.

Following this, in Figure 16y, a larger weight pushes down and the volume and pressure of the gas are measured again.

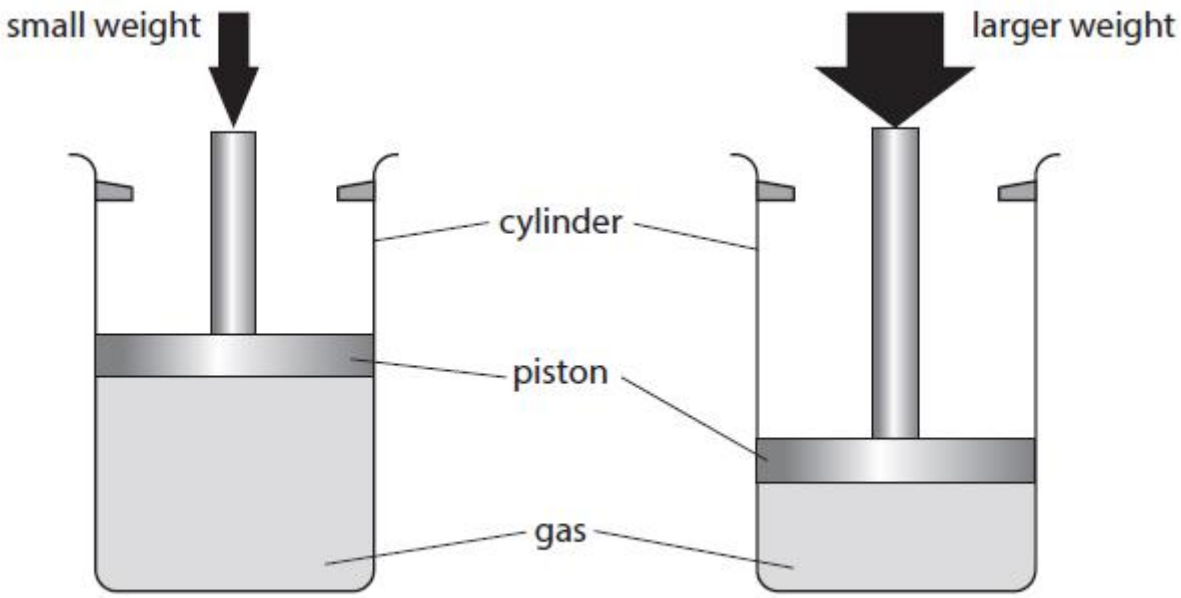

Figure 16x **Figure 16y**

Explain why the pressure and volume of the gas in the cylinder change when the larger weight pushes down on the piston.
The temperature of the gas does not change.

Your answer should include

- how the volume and pressure change
- how the gas particles exert a force on the piston

(6)

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(Total for question = 6 marks)

Q5.

A piece of wood has a similar shape and volume to the lump of iron.

The density of the wood is 0.82 g / cm^3.

The density of water is 1.00 g / cm^3

Explain why the method used in part (b) cannot be used to determine the mass of the piece of wood.

(2)

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(Total for question = 2 marks)

Q6.

Figure 23 gives information about the density of aluminium.

	density in g/cm^3
solid aluminium	2.70
liquid aluminium	2.38

Figure 23

Explain the difference between the density of solid aluminium and the density of liquid aluminium in terms of the arrangement of particles.

(2)

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(Total for question = 2 marks)

Q7.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

When the pressure in a container of gas increases, the particles of the gas

(1)

☐ **A** hit the sides of the container less often

☐ **B** hit the sides of the container more often

☐ **C** move with lower speeds

☐ **D** vibrate more energetically about fixed positions

(Total for question = 1 mark)

Q8.

When a diver is swimming under water, she breathes out bubbles of gas, as shown in Figure 27.

(Source: © mihtiander/123RF)

Figure 27

The bubbles of gas rise to the surface.
The temperature of the gas does not change.

Explain what happens to a bubble as it rises to the surface.
Your answer should refer to gas equations, kinetic theory and particles.

(6)

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(Total for question = 6 marks)

Q9.

Figure 6 shows helium gas inside a container.

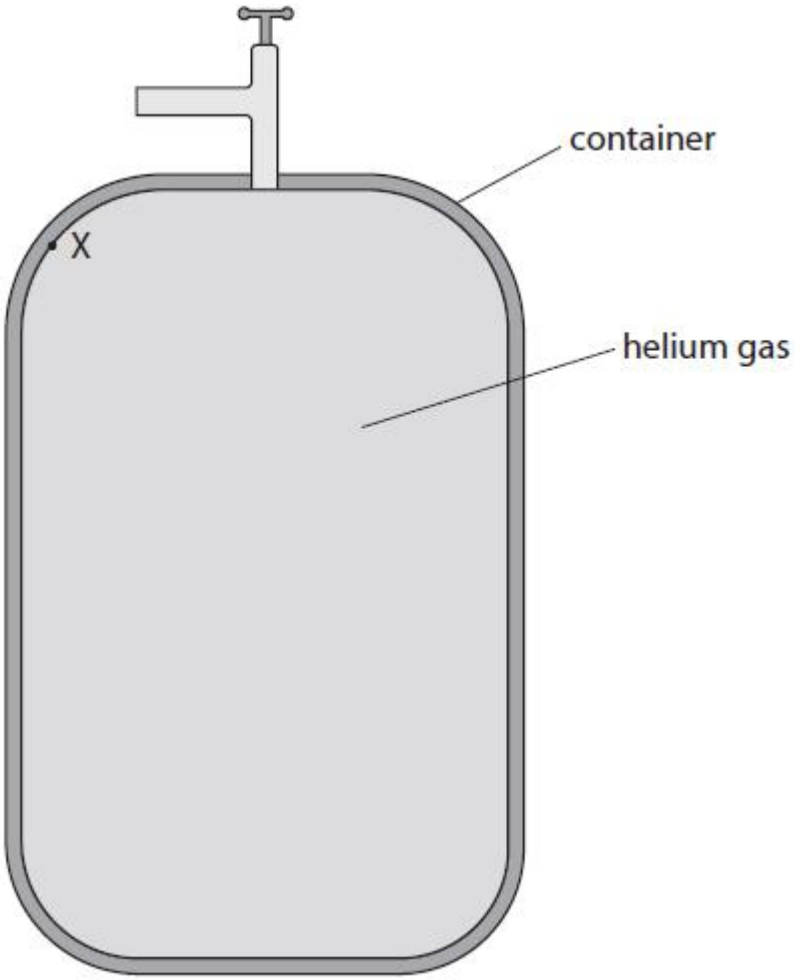

Figure 6

(i) Draw an arrow on Figure 6 to show the direction of the force due to the helium gas, at the point labelled X.

(1)

(ii) Explain, in terms of particles, why the helium gas exerts a force on the sides of the container.

(2)

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(Total for question = 3 marks)

Q10.

Figure 6 shows a graph produced from the data collected by the data logger in Figure 5.

Labels P, Q, R and S have been added.

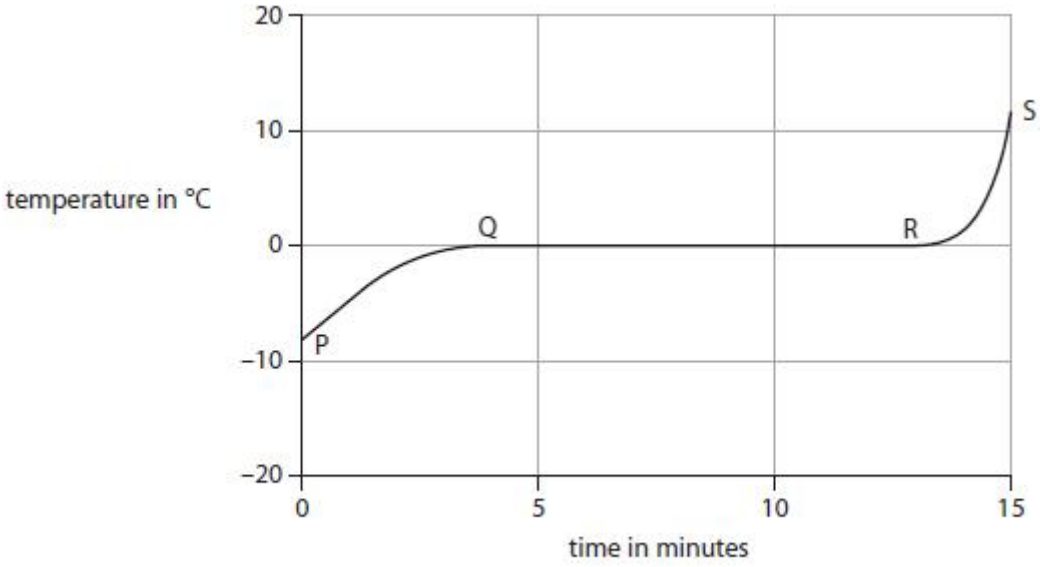

Figure 6

At the start, P, the beaker contains crushed ice at −8 °C.

Describe what happens to the crushed ice during the next 15 minutes.

You may use labels P, Q, R and S to help your answer.

(3)

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(Total for question = 3 marks)

Mark Scheme

Q1.

	Answer	Acceptable answers	Mark
(a)	☒ **C** (graph C)		**(1)**
(b)(i)	A description including: • collisions (1) • with (walls of) cylinder (1)	hit / bounce off exert force	**(2)**
(b)(ii)	substitution (1) either 100 × V = 15.0 × 21 000 or $V = \frac{15.0 \times 21\ 000}{100}$ evaluation (1) 3 150(litres)	V_1P_1 = 15 × 21000 = 315000 (1 mark) V_2P_2 = **100** × 3200 = 320000 (1 mark) award full marks for 3150 (litres) without working	**(2)**
(b)(iii)	substitution (1) $\frac{21\ 000\ (\times V)}{305} = \frac{P\ (\times V)}{278}$ volume same (1) evaluation (1) 19 100 (kPa)	give full marks for correct answer, no working transposition accept 19141 (kPa) or 19000 and numbers rounding down to 191 00	**(3)**

Q2.

Question number	Indicative content	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. • Gas particles are in (random) motion • Particles hit sides of container / each other • This produces (net) force on (sides of) container • This causes a pressure (on the container) • As volume increases the particles have more space to move • This means that particles hit sides less frequently • Smaller (net) force on sides of container • Pressure decreases. • Change in pressure (with increase in volume) is not linear • Pressure never becomes zero	(6)

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance simple description of how pressure is caused **or** how it changes	Possible candidate responses Particles hit sides of container **or** Pressure becomes less as volume increases
Level 2	3–4	Additional guidance description of how pressure is caused and how it changes	Possible candidate responses Particles hit sides of container. This causes a (net) force on sides of container. Pressure becomes less as volume increases
Level 3	5–6	Additional guidance Description of how pressure is caused, and why it changes in the way shown	Possible candidate responses Particles hit sides of container. This causes a (net) force on sides of the container. As volume increases the particles hit the sides less frequently so the pressure becomes less.

Q3.

Question	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. AO1 particles exerting pressure: • particles in (rapid) (random) motion/have high speed • particles move freely • particles collide • particles collide with the (walls of) the container • there is a change in momentum of the particles • producing a force on the walls • pressure = force/area OR P=F/A	**(6)** **AO1.1**
	effect of decreasing volume: • when volume decreases collisions increase • when volume decreases rate/frequency of collisions increases • force on walls increases so pressure increases • area of walls decreases • (P = F/A) (so) as area decreases pressure increases	

Level	Mark	Descriptor
	0	• No rewardable material.

Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	**Mark**	**Additional Guidance**	**General additional guidance – the decision within levels** e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material	
Level 1	1–2	Additional guidance isolated facts, statements from either section	Possible candidate responses particles are moving
Level 2	3–4	Additional guidance limited explanation of both sections OR detailed explanation of one section	Possible candidate responses particles are moving and collide more particles in a gas are moving freely colliding with the walls of the container

Level 3	5–6	Additional guidance detailed explanation from both sections	Possible candidate responses particles in a gas are moving freely, colliding with the walls of the container as the volume decreases (rate of) collisions increases

Q4.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive, and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. AO2.1/ AO3 **How volume and pressure change** • the weights on the piston compress the gas (by applying a force) • larger weight causes the volume to decrease • pressure increases when the volume decreases • (in a smaller volume) particles collide more frequently (with surface)	**(6)** **AO2.2, AO3.3a**

	• (more frequent collisions) causes a gas particles to exert a greater force (on container) • P=F/A **How particles exert force** • gas particles collide with piston (sides of container) • there is a change in momentum of the gas particles • force equals rate of change of momentum • force is exerted over an area causing pressure • force exerted by weight is balanced by force exerted by gas	

Level	Mark	Descriptor
	0	• No awardable content
Level 1	1–2	• Interpretation and evaluation of the information attempted but will be limited with a focus on mainly just one variable (AO3) • The explanation attempts to link and apply knowledge and understanding to scientific ideas, flawed or simplistic connections made between elements in the context of the question.(A02)
Level 2	3–4	• Interpretation and evaluation of the information on both variables synthesises mostly relevant understanding (AO3) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas some logical connections made between elements in the context of the question(AO2)
Level 3	5–6	• Interpretation and evaluation of the information, demonstrating throughout the skills of synthesising relevant understanding (AO3) • Then explanation is supported is supported throughout by linkage and application of knowledge and understanding of scientific ideas some logical connections made between elements in the context of the question (AO2)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Isolated fact about volume, pressure, movement of gas particles	Possible candidate responses smaller volume (of gas) pressure is increased gas particles collide
Level 2	3–4	Additional guidance Limited explanation linking weight/force with pressure and volume OR linking movement of gas particles/ collisions (with walls) to force exerted by gas	Possible candidate responses the volume gets smaller, and the pressure goes up OR gas particles collide causing a force gas particles have more frequent collisions
Level 3	5–6	Additional guidance Detailed explanation that includes linking weight/force with pressure and volume and linking collisions to force or frequency of collisions	Possible candidate responses a larger weight gives a greater pressure and a smaller volume, the gas particles exert a force through collisions in Y, the volume is smaller, and the pressure is bigger and the particles have more frequent collisions

Q5.

Question number	Answer	Additional guidance	Mark
	an explanation linking density of wood less (than that of water) (1) less (volume of) water displaced (than volume of wood) (1)	 allow wood floats / should be submerged allow wood absorbing water allow (idea of) incorrect volume reading allow (idea that) the volume cannot be measured this way	**(2)** **A02**

Q6.

Question	Answer	Additional guidance	Mark
	an explanation linking: density of solid is greater (than density of liquid) (1) (because) distance between particles in solid is less (than distance between particles in liquid) (1)	 solids are dens**er** accept in solids, particles are clos**er** accept in solids, there are **more** particles per unit volume / particles are **more** (tightly) packed	**(2)** **AO1.1**

Q7.

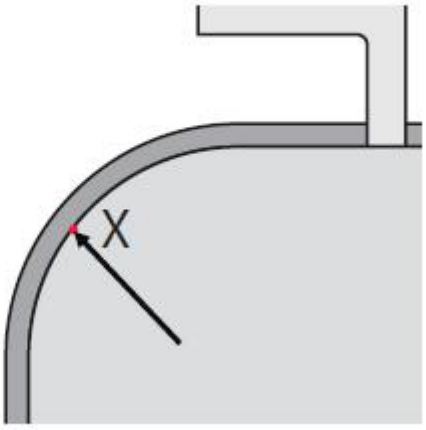
Question number	Answer	Mark
	B hit the sides of the container more often Options A, C and D are incorrect associations	(1) AO1

Q8.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO2 (6 marks)** • The bubbles get bigger • Molecules of gas in constant motion • Molecules widely spaced and moving randomly • Molecules impact on surface of bubble/liquid molecules • Average of impacts produces gas pressure • Pressure is due to rate at which gas particles collide with liquid molecules/bubble surface • Liquid pressure decreases as bubble rises • $P_1V_1 = P_2V_2$ • If pressure decreases, volume of bubble will increase/volume of bubble must increase to give a decrease in pressure • As volume increases, rate at which particles collide with surface of bubble decreases	(6)

Level	Mark	Descriptor
	0	No awardable content.
Level 1	1–2	The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2) Lines of reasoning are unsupported or unclear. (AO2)
Level 2	3–4	The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2) Lines of reasoning mostly supported through the application of relevant evidence. (AO2)
Level 3	5–6	The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2) Lines of reasoning are supported by sustained application of relevant evidence. (AO2)

Q9.

Question number	Answer	Additional guidance	Mark
(i) **Clip with fig 6**	arrow normal to surface at point x 	Judge by eye may be inside or outside the container	**(1)** **AO1**

Question number	Answer	Additional guidance	Mark
(ii)	particles are in (random) motion (1) collide with sides (of container) (1)	 particles bounce off sides of container award 2 marks for change in momentum of particles.	(2) AO1

Q10.

Question number	Answer	Additional guidance	Mark
	A description to include any three from from P to Q temperature (of ice) increases (1) from Q to R temperature (of ice and water mixture) stays constant (at 0°C) (1) from R to S temperature (of water) increases (1) any reference to melting / melts (1)	accept time interval references as equivalent to PQ, QR and RS intervals take PQ as from -8/-9°C to Q take RS as from R to to 11/12 °C accept solid/ice turns to liquid/water	(3) AO3

	melting from Q to R (1) PQ (contents are) solid (1) QR (contents are) {solid + liquid} / {ice + water} (1) RS (contents are) liquid / water (1)	temperature stays constant when melting if no other mark scored, allow one mark for correct description of temperature changes without references to PQRS or time	